

The Correlation Between Survival Rates in Himalayan Mountain Expeditions and Climbers' Economic Backgrounds

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1 Introduction

The high-altitude peaks of the Himalayas present some of the most extreme physical and environmental challenges on Earth. While individual physiological fitness, age, and technical mountaineering skills are widely recognized as critical factors in expedition outcomes, this project explores a systemic socioeconomic variable: the climber's macroeconomic background.

This project investigates whether a climber's national economic status (measured by looking at the national GDP per capita) makes a huge influence on survival rates during Himalayan expeditions or not. The primary hypothesis shows that higher national wealth correlates with superior baseline training, higher-quality mountaineering gear, and comprehensive institutional rescue or medical infrastructure, therefore increasing survival probabilities under extreme conditions.

2 Data Overview

To find out the correlations and predictive interactions among demographics, geography, and macroeconomic background, I integrated three distinct public data sources into a unified data pipeline:

- **Climber Demographic & Expedition Data (`members.csv`):** This dataset shows historical tracking of over 76,000 individual climbers, containing records of age, sex, citizenship, season of climb, deployment of supplemental oxygen, and survival status (dead/alive).
- **Macroeconomic Indicator Data (World Bank GDP Dataset):** Shows the recordings of the historical “GDP per capita (current US\$)” across global nations from 1960 to the near present.
- **Topographical Peak Data (`peaks.csv`):** This geographic data details specific traits of targeted Himalayan mountains, primarily focusing on absolute altitude scaled in meters (`height_metres`).

3 Data Cleaning and Preparation

The raw data that been collected such as demographic registries, geographic logs and economical table needed to be formatted in the right way and ensured that aligned in the correct column/row format to get the true results.

3.1 Climber Registry Processing

The initial climber dataset was loaded from its raw file. Irrelevant columns were dropped, retaining only indicators that we need: `citizenship`, `year`, `age`, `sex`, `oxygen_used`, `season`, `peak_id`, and fatality status. Categorical values like sexes were mapped into a standardized space (`sex_enc`), while boolean vectors such as oxygen usage were explicitly coerced to numeric binary flags (0 for unused, 1 for used).

3.2 Macroeconomic Data Reshaping

The World Bank GDP dataset was structured in a wide format where individual years represented distinct columns. The attributes which we don't use such as (`Country Code`, `Indicator Name`, `Indicator Code`, and trailing unallocated dimensions) are stripped. I applied a pandas `melt` operation to convert the dataset into a long-form structure indexed uniquely by `Country Name` and `year`, ensuring compatibility with the temporal granularity of the expedition data. Values inside the `year` column were converted to numerical integers, and missing values in the economic arrays were deleted.

3.3 Mountain Topography Merging

The mountain data file (`peaks.csv`) was filtered to extract absolute altitude variables. Using the `peak_id`, the absolute mountain height (`height_metres`) was left-joined into the main data structure. The final combined data frame dropped rows containing null records in critical fields to guarantee an uncorrupted matrix for statistical validation and model training.

4 Exploratory Data Analysis

For the exploratory data analysis I focused on mapping the demographic attributes of the sample population. I made line profiles, count plots, and frequency histograms to review the base of the data distributions.

Initial plotting shows a huge gender imbalance across historical records, with male climbers constituting roughly 95% of the total dataset. Age distributions revealed a standard bell-curve profile centered within the 25–45 age range. Additionally, calculating the overall historical baseline fatality rate revealed that deaths are highly rare anomalies, that only $\sim 1.3\%$ percent of the recorded datasets accounts this aspect.

Figure 1 shows the top 15 countries by total number of registered Himalayan climbers, with Nepal leading by a large margin reflecting the integral role of local climbers and Sherpas in expedition operations.

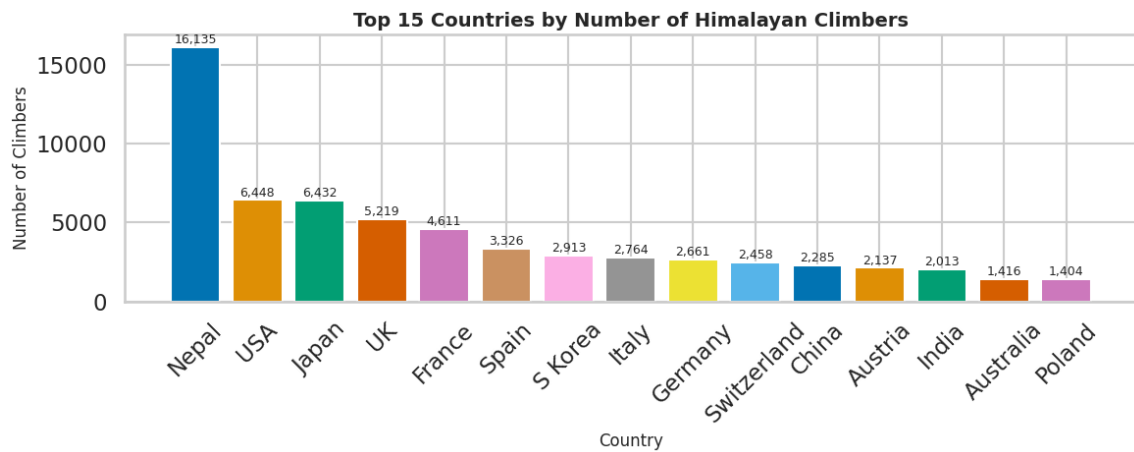


Figure 1: Top 15 Countries by Number of Himalayan Climbers.

Figure 2 presents the fatality rate per citizenship for the top 15 most represented countries. India, Poland, and South Korea show the highest fatality rates (above 1.99%), while the USA, China, and the UK remain well below the overall average of 1.45%.

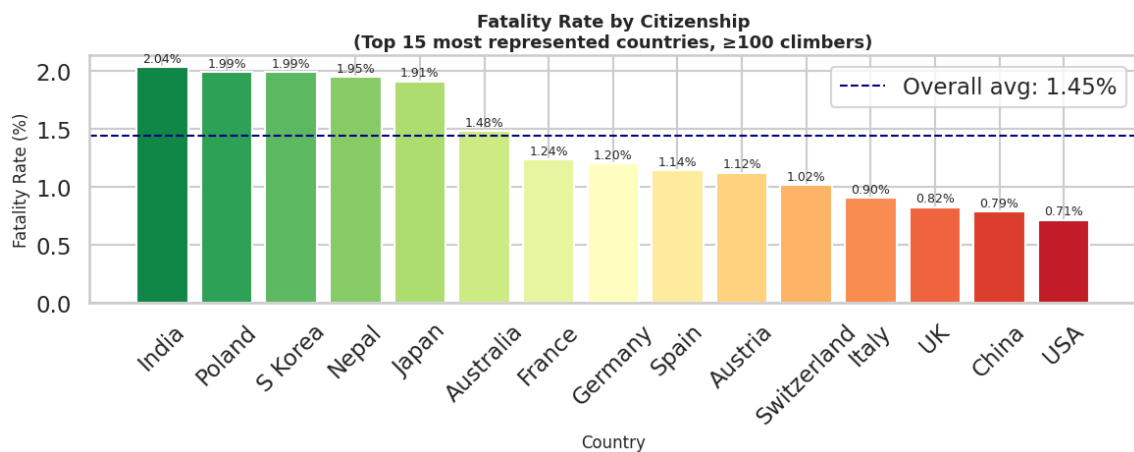


Figure 2: Fatality Rate by Citizenship (Top 15 most represented countries, ≥ 100 climbers) — vertical bar view.

Figure 3 illustrates the age distributions of climbers who survived versus those who died. The distributions largely overlap, with the deceased group showing a slightly younger mean age (36.0) compared to survivors (37.4).

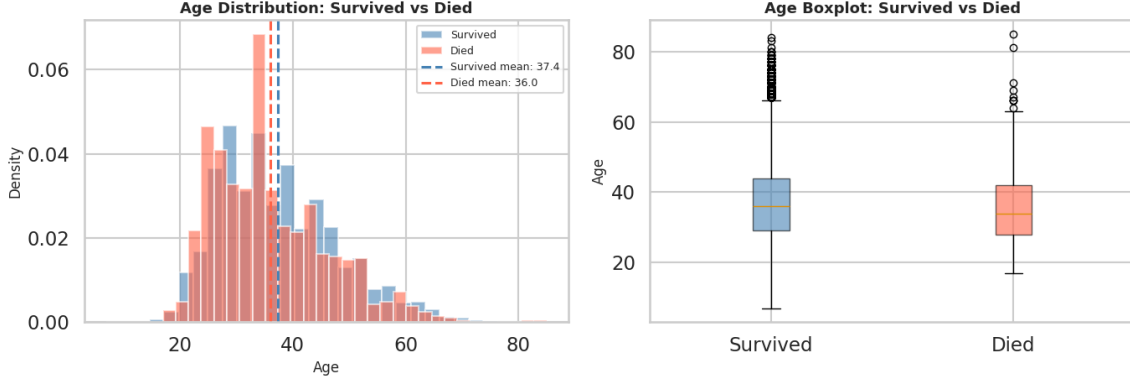


Figure 3: Age Distribution and Boxplot: Survived vs. Died.

Figure 4 shows fatality rates across the four climbing seasons. Winter exhibits the highest fatality rate at 2.24%, nearly double that of Autumn (1.37%), the safest season.

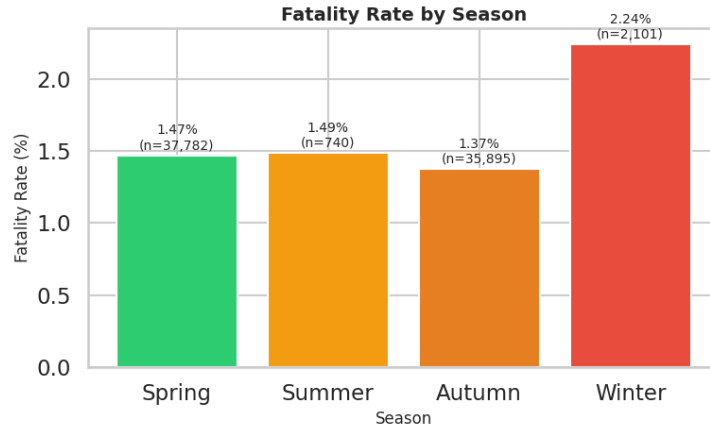


Figure 4: Fatality Rate by Season.

5 Statistical Analysis and Hypothesis Testing

For evaluating the baseline relationships between individual attributes, economic indicators, and survival outcomes, I conducted separate statistical tests.

- **Hypothesis 1 (GDP vs. Survival):** I conducted an Independent T-Test to compare mean national GDP per capita between surviving and non-surviving populations. The test gave us a highly significant result ($p < 0.00001$), which confirms a strong mathematical association where climbers from higher GDP backgrounds experience lower relative fatality rates.
- **Hypothesis 2 (Gender vs. Survival):** Using a Chi-Square test of independence that we applied to the sex and fatality probability arrays verified a statistically significant result ($p = 0.00011$), showing that raw survival ratios can vary a lot between gender classifications.
- **Hypothesis 3 (Citizenship vs. Survival):** To minimize unexpected outcome from the minor samples, I filtered the dataset to include only countries with ≥ 200 registered

climbers. After that a Chi-Square test applied to this subset confirmed a significant relationship ($p < 0.05$) between citizenship and survival, demonstrating that survival odds are not distributed equally across national cultures.

Figure 5 provides a horizontal view of fatality rates by citizenship, highlighting how nations above and below the 1.45% overall average differ markedly.

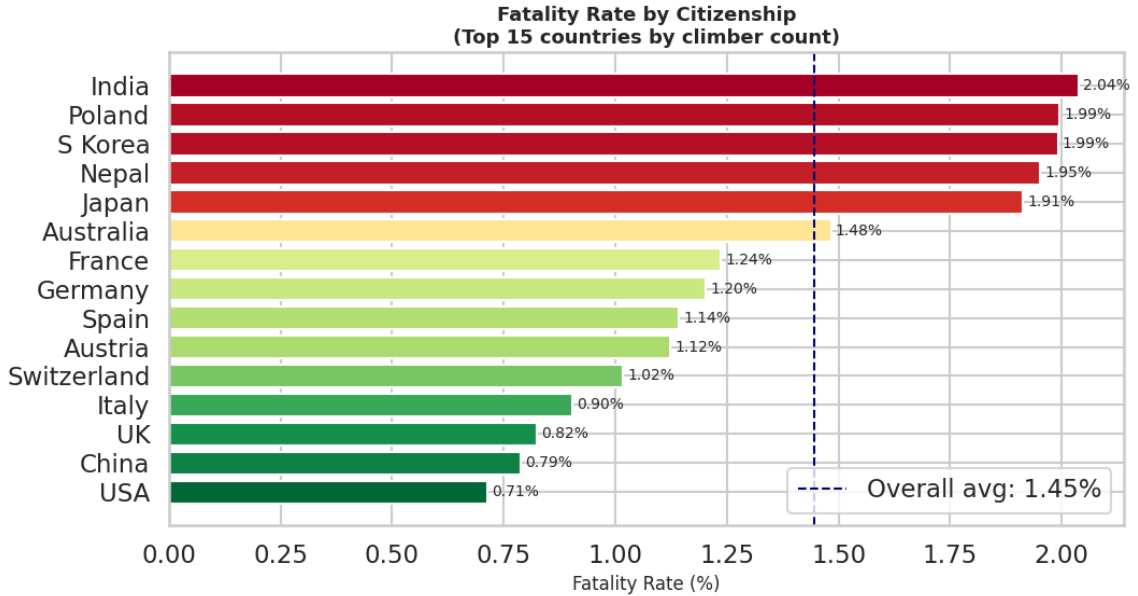


Figure 5: Fatality Rate by Citizenship — Top 15 Countries by Climber Count (horizontal bar view).

6 Machine Learning: Random Forest Classification

While statistical tests confirmed that individual correlations exist, they evaluate variables in isolation. To understand how demographic factors interact simultaneously with geographic and physical constraints together, I implemented a machine learning model.

6.1 Model Architecture

For the predictive part I utilized a Random Forest Classifier for optimizing an unprocessed array of continuous and encoded parameters into a processed one:

- **Input Features:** age, gdp_per_capita, sex_enc, oxygen_enc, season_enc, height_metres.
- **Target Array:** Binary mortality status (died: 0 for survived, 1 for deceased).

The data was divided into 2 as 80% training set and 20% for the test set using stratified sampling to preserve the rare mortality ratio across both splits. To prevent the model from ignoring the minority class due to the extreme 99:1 survival imbalance, the algorithm is configured with `class_weight='balanced'`, which automatically adjusts the penalty function to place higher emphasis on correctly identifying fatality events. The ensemble was constructed utilizing 200 distinct decision estimators (`n_estimators=200`).

6.2 Evaluation and Visualization

Model performance was evaluated by a classification matrix, tracking precision, recalls, and F1 metrics across both outcome classes. Given the class imbalance, performance was evaluated using the Area Under the Receiver Operating Characteristic curve (ROC-AUC). The model achieved a stable ROC-AUC score of 0.71, confirming meaningful predictive capacity well above a random guessing threshold (0.5).

Figure 6 shows the confusion matrix for the Random Forest model on the held-out test set.

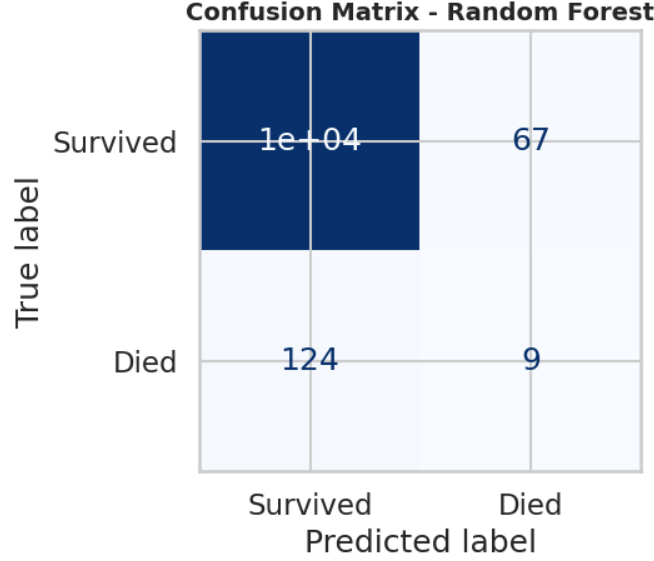


Figure 6: Confusion Matrix — Random Forest Classifier.

6.3 Experiments and Findings

By extracting the feature importance from the ensemble, the machine learning model allowed us to rank the predictive weight of each variable (Figure 7):

- **GDP per Capita (~ 44%):** Found as the biggest indicator. Confirming that fatality rate is heavily dependent on the macroeconomic background over long historical scales.
- **Age (~ 26%):** Ranked second, captures how physical endurance naturally declines at extreme altitude by getting older.
- **Peak Height (~ 22%):** Represented the primary environmental threat, as ultra-high altitude peaks exponentially increase hypoxia risks.
- **Season (~ 4%), Oxygen Used (~ 2%), Gender (~ 2%):** Exhibited lower standalone importance inside the model. For instance, oxygen use correlates tightly with extreme peak heights, meaning its predictive signal was largely absorbed by the `height_metres` feature.

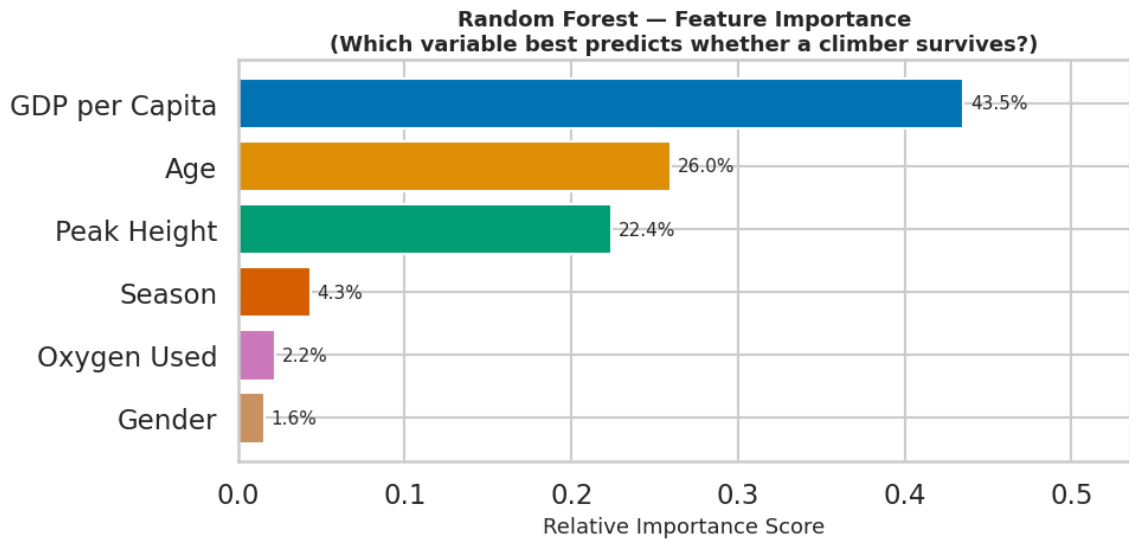


Figure 7: Random Forest — Feature Importance (relative importance scores per variable).

Combining the Chi-Square citizenship results with the ML findings added a critical layer of real-world results. Nations such as Nepal and India display higher relative fatality rates because their majority of the populations predominantly take on high-risk, high-exposure operational roles on the mountain (e.g., high-altitude porters, rope-fixers, and Sherpas). On the other hand, western nations (e.g., USA, UK) have lower fatality metrics, reflecting the protective benefits of premium guided packages and extensive support systems.

7 Conclusion

The integration of traditional statistical validation alongside machine learning modeling provided clear insights into Himalayan expedition safety. As we see that physiological age and absolute geographical scale affects majorly in the fatality rate; we also prove that macroeconomic background (GDP per capita) acts as an essential variable that significantly influences the survival odds.

However, these findings should be interpreted with an understanding of our data limitations. The extreme rarity of fatalities ($\sim 1.3\%$) creates a severe statistical imbalance that limits overall classification precision. Despite these data constraints, our modeling approach proved highly effective for ranking variables and understanding how economic context shapes survival outcomes in extreme environments.